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--- RAG METADATA ---
document_source: MoHFW_Patient_Safety_Guidelines_1st_Edition
clinical_domain: risk_assessment_matrix
target_demographic: general_triage
chapter_pages: 11-17
safety_critical: true
rag_index: risk_management_process
keywords: risk management, risk assessment, risk register, likelihood, impact, red zone, yellow zone, green zone
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Risk Management Process in Hospital

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Risk Assessment

Risks are assessed using a qualitative two-fold criteria: (a) the likelihood of occurrence of the risk event and (b) the magnitude of impact if the risk event occurs. Their combination produces the inherent risk level. Both likelihood and impact are rated over a period of 12 to 18 months, assessed in the context of existing controls. Where statistical data is unavailable, subjective estimates reflecting employee or group belief are used.

Likelihood Scale (1–5)

- 5 — Very high likelihood
- 4 — High likelihood
- 3 — Moderate likelihood
- 2 — Low likelihood
- 1 — Very low likelihood

Impact Scale (1–5)

- 5 — Very high impact
- 4 — High impact
- 3 — Moderate impact
- 2 — Low impact
- 1 — Very low impact

Hospital Process Risk Domains

The following domains represent common hospital processes, their associated threat profiles, and mitigation requirements. Each domain should be scored using the Likelihood x Impact matrix to determine risk priority.

Domain 1: Quality Management System

- Threat Profile: Systemic non-conformity leading to holding or withdrawal of certificate of Accreditation or License.
- Risk Areas: Infection control quality, patient care service quality, physician licensing and credentialing, HMIS documentation and reporting, clinical standards and practices, emergency procedures, and clinical audits.
- Mitigation Requirement: Establish and maintain a robust quality management system with regular internal audits and accreditation compliance tracking.

Domain 2: Health and Safety

- Threat Profile: Risks associated with environment pollution, safety of resources, and employees' health and security at healthcare establishments.
- Mitigation Requirement: Enforce occupational health and safety standards; conduct regular environmental safety assessments.

Domain 3: Medical Services and Patient Healthcare Fulfilment

- Threat Profile: Medication errors, surgical errors, other procedure errors, clinical negligence, documentation errors, injury to patient or healthcare service provider, patient falls, hospital-acquired infections.
- Risk Areas: Risks related to multidisciplinary approach to acute care, specialty care, diagnostics, investigations, and wellness programs — including risks from inadequate facilities and inaccurate treatment across all service areas.
- Mitigation Requirement: Implement medication reconciliation protocols, surgical safety checklists (WHO SSC), fall prevention programs, and HAI surveillance.

Domain 4: Service Excellence

- Threat Profile: Risks associated with inadequate infrastructure to support patient services, patient satisfaction, and care for inpatients, outpatients, and referred patients.
- Mitigation Requirement: Regular patient satisfaction surveys; ensure adequate infrastructure and capacity planning for all patient categories.

Domain 5: Nursing Operations

- Threat Profile: Risks related to the adequacy of policies and procedures governing nursing operations and maintaining continuity of care.
- Mitigation Requirement: Maintain up-to-date nursing SOPs; conduct regular competency assessments and handover audits.

Domain 6: Registration and Admission of Patients

- Threat Profile: Documentation errors, delays in admission processing, and other administrative errors.
- Mitigation Requirement: Standardise registration workflows; implement electronic pre-registration and real-time queue management.

Domain 7: Medical Records Management

- Threat Profile: Documentation errors, unauthorized access to patient records, loss or damage of records, fire hazard.
- Mitigation Requirement: Implement role-based access control on EMR systems; maintain physical record backup protocols and fire-safe storage.

Domain 8: Radiology and Imaging

- Threat Profile: Imaging errors, errors in interpretation of images, documentation errors, drug reactions during contrast administration.
- Mitigation Requirement: Enforce double-reporting protocols for critical imaging; maintain pre-contrast allergy screening checklists.

Domain 9: Laboratory

- Threat Profile: Test result errors, internal documentation mistakes, uncalibrated equipment.
- Mitigation Requirement: Regular equipment calibration schedules; conduct vendor evaluation for reagent suppliers; implement delta-check protocols for result verification.

Domain 10: Purchase, Receipt, Inventory Control, and Disposal

- Threat Profile: Ineffective purchasing (wrong material quality, excess or short quantity, inflated price, delayed delivery), high inventory levels, damage, deterioration or pilferage, stock-outs.
- Mitigation Requirement: Implement inventory management systems with reorder-point alerts; conduct periodic vendor performance reviews and stock audits.

Domain 11: Control of Outsourced Processes

- Threat Profile: Ambiguity in defining scope of services, ambiguity in commercial terms and conditions, unrealistic contract pricing, non-conformity to legal requirements in contract documents.
- Mitigation Requirement: Use standardised service-level agreements (SLAs); conduct pre-contract legal review and periodic SLA compliance audits.

Domain 12: Housekeeping

- Threat Profile: Non-conformity to Bio-Medical Waste Management Rules, non-compliance with infection control protocols, inadequate sterilization of facilities including OT, Labour Room, ICU, SNCU, and Wards.
- Mitigation Requirement: Enforce BMW management compliance per CPCB guidelines; conduct regular sterilization audits of all critical areas.

Domain 13: Training of Staff

- Threat Profile: Lack of resources for training, lack of interest from staff and management.
- Mitigation Requirement: Develop annual training calendars; make mandatory training completion a KPI for all clinical and non-clinical staff.

Domain 14: Maintenance

- Threat Profile: Lack of resources for preventive maintenance, calibration and repair; lack of opportunity to release equipment for preventive maintenance and calibration.
- Mitigation Requirement: Implement a Computerised Maintenance Management System (CMMS); schedule equipment downtime windows in coordination with clinical teams.

Domain 15: Supply of Therapeutic Diet

- Threat Profile: Patient or family party non-compliance with therapeutic diet advice.
- Mitigation Requirement: Assign dedicated clinical dietitians; reinforce dietary education during admission and at discharge through illustrated materials.

Domain 16: Pharmacy

- Threat Profile: Risks associated with operation of pharmacy and delivery of pharmaceutical products to hospital wards, critical care areas, and outpatients.
- Mitigation Requirement: Implement unit-dose dispensing systems; conduct regular pharmacy audits and narcotic/high-alert drug reconciliation checks.

Domain 17: Human Resource

- Threat Profile: Risks associated with organisational culture, structure, communication, recruitment, performance management, remuneration, learning and development, retention, Occupational Health and Safety, and institutional relations — including supporting systems, processes, and procedures.
- Mitigation Requirement: Implement structured HR policies, grievance redressal mechanisms, and regular staff engagement surveys.

Domain 18: Information Technology

- **Threat Profile:** Inadequate management or control of IT systems, failure to ensure data integrity and reliability, inadequate HMIS/HIS performance and monitoring, system or network architecture not supporting medium- or long-term business strategy, capacity planning not reviewed regularly resulting in processing failures, risks of data or systems migration or interface failures.
 - **Mitigation Requirement:** Establish IT governance frameworks; conduct regular DR (Disaster Recovery) drills, data backups, and HMIS performance audits.
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Risk Evaluation

Impact and Likelihood scores are multiplied to produce a composite risk score. The average score across the assessor group should be calculated. Risks are then classified into one of three zones:

- **RED Zone (Score 15–25): CRITICAL** — Requires immediate action plans to close significant control gaps. Needs active and continuous monitoring.
- **YELLOW Zone (Score 8–12): CAUTIONARY** — Action steps to develop or enhance existing controls are required. Needs periodic monitoring.
- **GREEN Zone (Score 1–6): ACCEPTABLE** — Risk is considered in control. Needs annual review only.

Risk Score Matrix (Likelihood x Impact)

Likelihood \ Impact	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

Format for Risk Register (Key Fields)

- Date of Identification
- Risk Title
- Risk Description
- Possible Cause
- Likelihood of Risk (A) — scale 1 to 5
- Severity of Risk (B) — scale 1 to 5
- Impact Factor (A x B) — computed risk score
- Risk Category
- Risk Owner
- Date of Assessment
- Date of Resolve
- Residual Risk
- Date of Surveillance
- Risk Avoidance / Mitigation Strategy